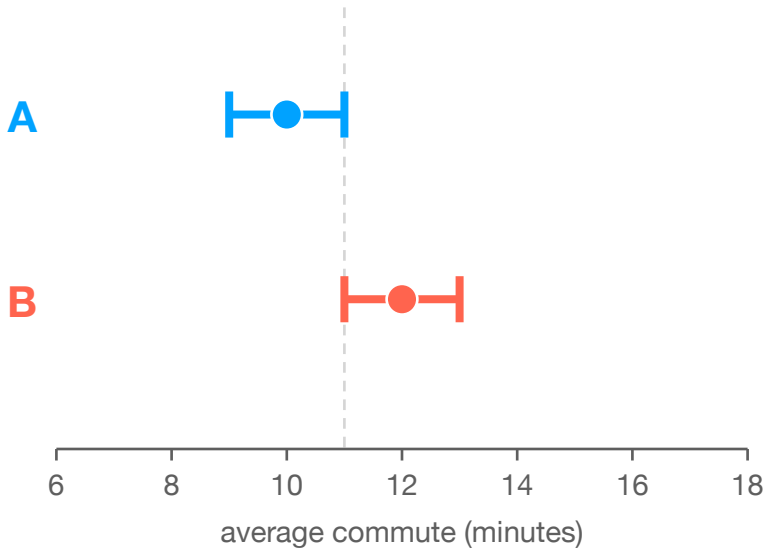
# Reading uncertainty plots

**True or false? Error bars, confidence intervals, and sampling distributions**

Each plot gets its story first, then statements to mark T or F, then the answers. Throughout: estimates are approximately normal, SEs are known, and “significant” means at $\alpha = 0.05$.

# ± 1 standard error bars

Route A vs route B: average of 25 commutes each. Bars: average ± 1 SE.




## The question

Is bus route B slower than route A?

## The data

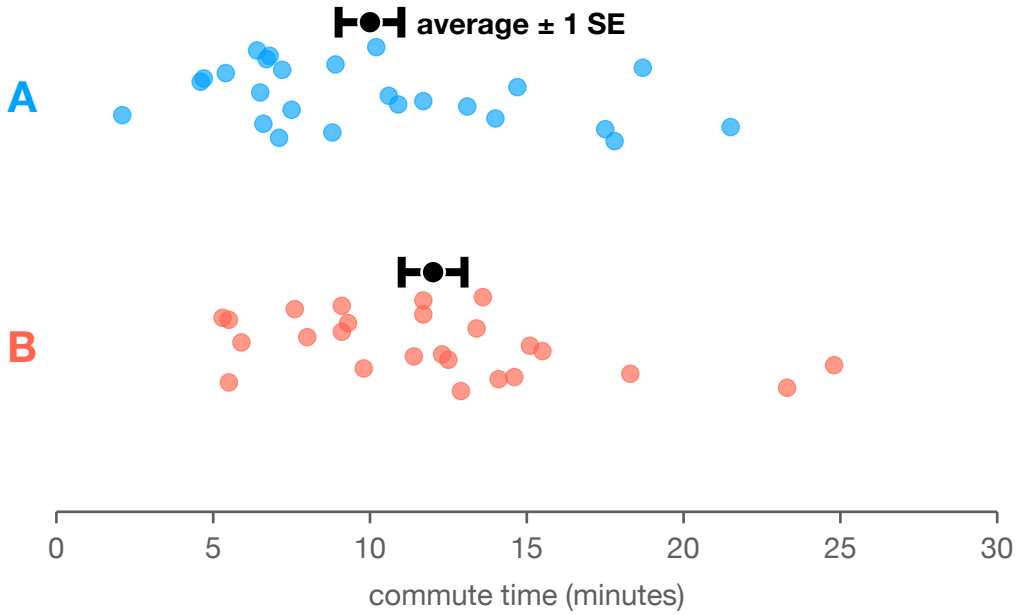
You time 25 commutes on each route. Route A averages 10 minutes and route B 12. Single trips vary a lot: SD = 5 minutes.

## The plot

Each dot is a route's average. Each bar is ± 1 standard error, SE = $5/\sqrt{25}$ = 1 minute: about how far that average would move if you timed 25 new trips.

# The data behind the plot

± 1 standard error bars




## Each dot

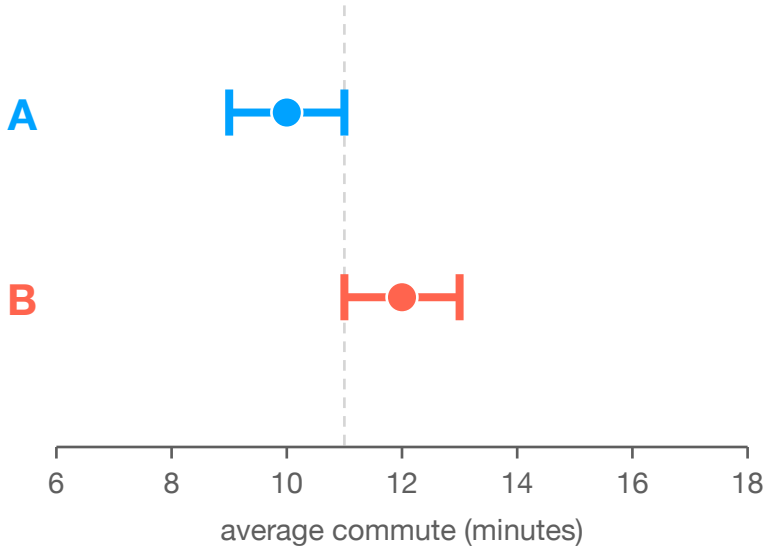
One timed trip: 25 per route.

## The black bar

The average ± 1 SE. It is much narrower than the dots, because it describes the average of 25 trips, not a single trip.

# ± 1 standard error bars

Route A vs route B: average of 25 commutes each. Bars: average ± 1 SE.




1 The bars don't overlap, so the difference is significant. T / F

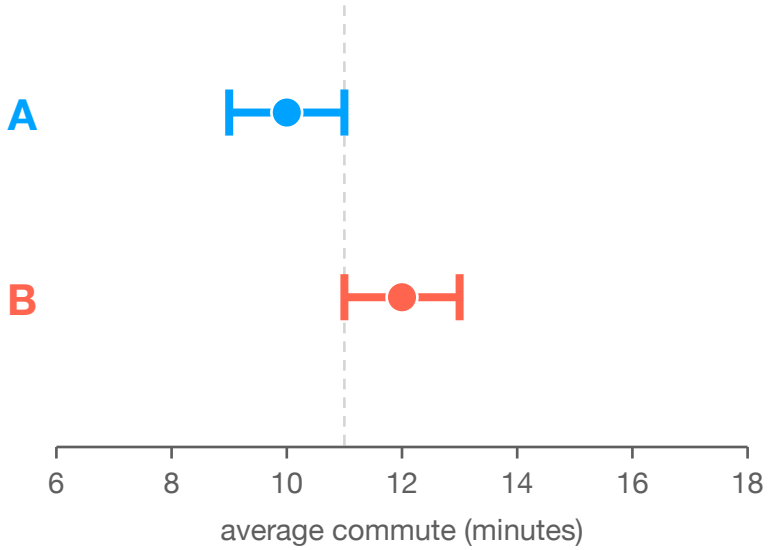
2 If we redid the study many times, about 68% of bars built this way would cover the route's true average. T / F

3 There's a 68% chance route A's true average is between 9 and 11 minutes. T / F

4 The data show that route B is slower on average than route A. T / F

# ± 1 standard error bars

Route A vs route B: average of 25 commutes each. Bars: average ± 1 SE.




## Answers

**1** The bars don't overlap, so the difference is significant. **FALSE**

$z = (2 - 0)/\sqrt{(1^2 + 1^2)} = 2/1.41 = 1.41$, so $p \approx 0.16$.

**2** If we redid the study many times, about 68% of bars built this way would cover the route's true average. **TRUE**

±1 SE covers about 68% of a normal curve.

**3** There's a 68% chance route A's true average is between 9 and 11 minutes. **FALSE**

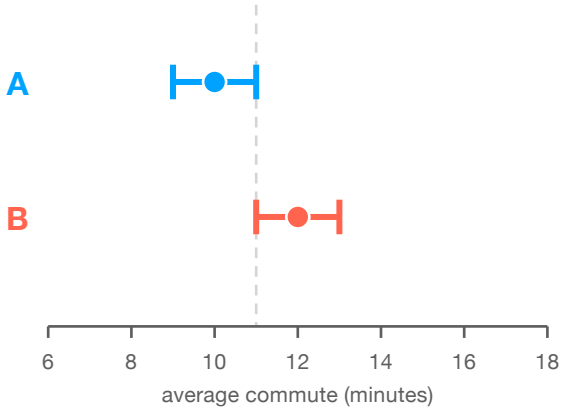
The 68% describes the procedure. (A flat-prior Bayesian would say 68%.)

**4** The data show that route B is slower on average than route A. **FALSE**

B's sample average is higher (12 vs 10), but that gap isn't significant: $p \approx 0.16$ (one-sided $p \approx 0.08$).

# Where it goes wrong

± 1 standard error bars




**1. “The bars don’t overlap, so the difference is significant.”**

**A/B testing:** A team sees the new checkout page’s ±1 SE bar sit clear of the old one and ships it. A gap this size turns up about 1 time in 6 when the two pages perform identically, so many of these “wins” vanish after launch.

**3. “There’s a 68% chance route A’s true average is between 9 and 11 minutes.”**

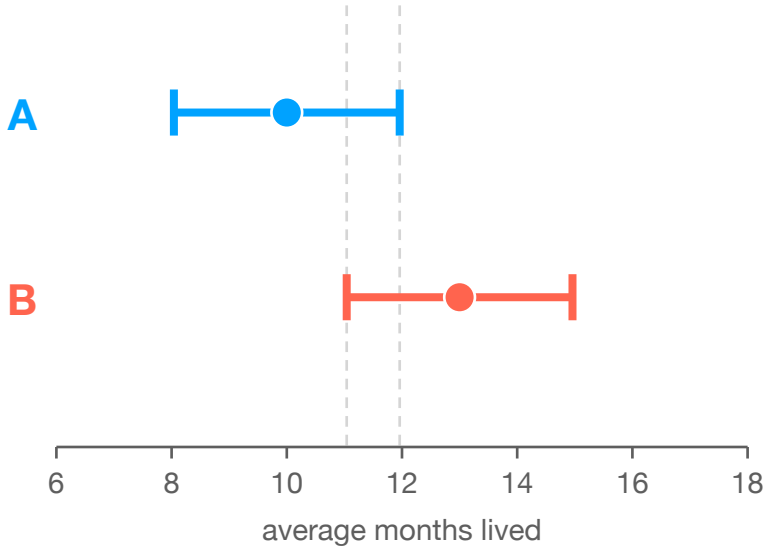
**Base-rate neglect:** A precognition study reports a 95% CI that excludes zero, and readers say they’re “95% sure” it’s real. How sure you should be also depends on how plausible the claim was beforehand: the base rate the CI ignores.

**4. “The data show that route B is slower on average than route A.”**

**Rankings:** League tables rank hospitals or schools by point estimate, and headlines say “B outperforms A”. With a gap this small relative to the SEs, next year’s data can easily flip the order.

# ± 95% confidence intervals

Standard drug (A) vs new drug (B): average months lived, 25 patients each. Bars: 95% CI.




## The question

Do patients on a new cancer drug (B) live longer than patients on the standard drug (A)?

## The data

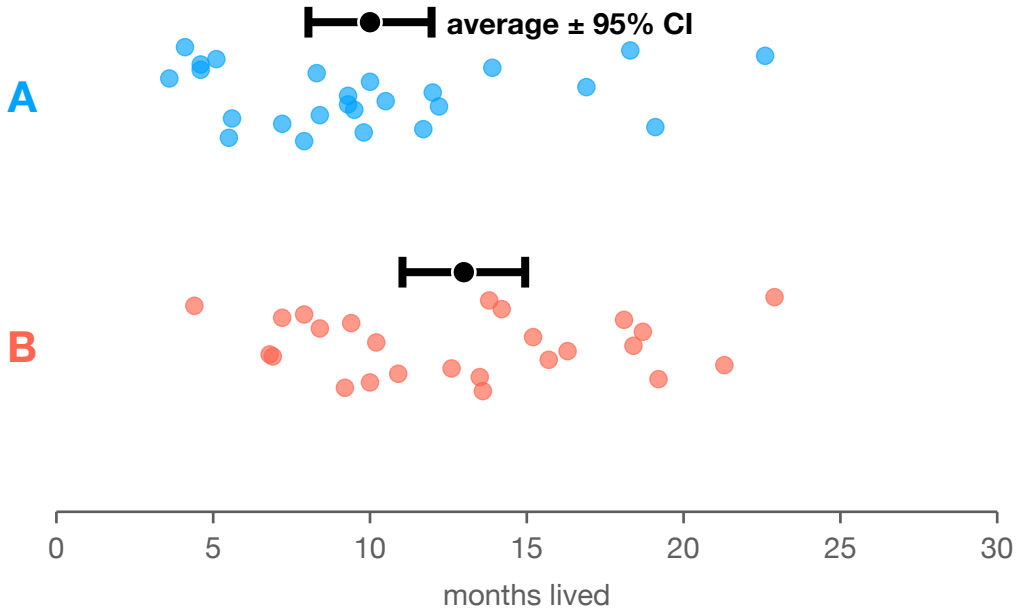
25 patients per drug. Patients on A live an average of 10 months after treatment and patients on B 13. Survival varies a lot: SD = 5 months, so each SE = $5/\sqrt{25} = 1$.

## The plot

Each bar is a 95% CI: average ± 1.96 SE, about ± 2 months. These bars are about twice as long as SE bars, so they overlap more easily.

# The data behind the plot

± 95% confidence intervals




## Each dot

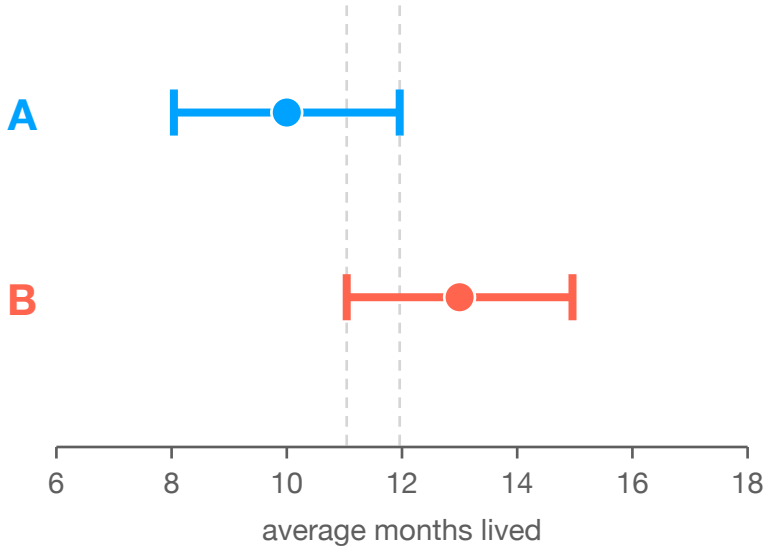
One patient: 25 per drug.

## The shape

Survival is skewed: a few patients live much longer. Averages of 25 patients are still close to normal (the CLT), which is what the CI relies on.

# ± 95% confidence intervals

Standard drug (A) vs new drug (B): average months lived, 25 patients each. Bars: 95% CI.



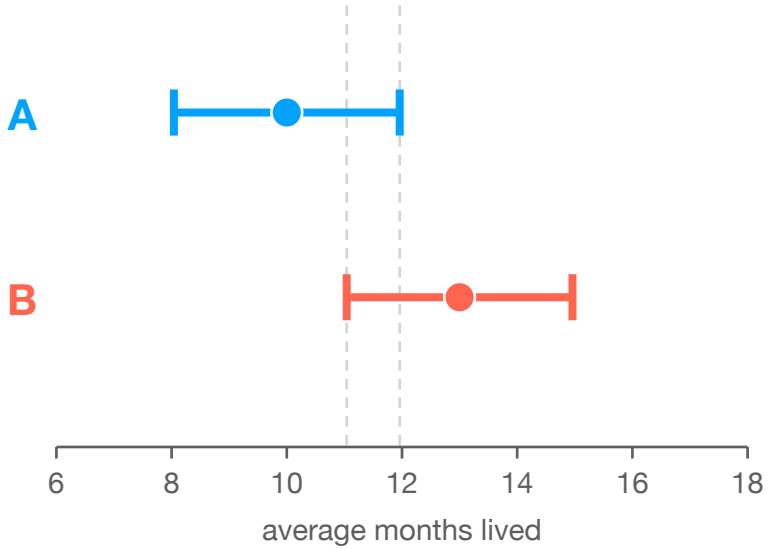

1 The CIs overlap, so the difference isn't significant. T / F

2 The 95% CI for the difference B – A is [0.23, 5.77], which excludes 0. T / F

3 If two 95% CIs don't overlap at all, the difference is significant. T / F

# ± 95% confidence intervals

Standard drug (A) vs new drug (B): average months lived, 25 patients each. Bars: 95% CI.




## Answers

**1** The CIs overlap, so the difference isn't significant. **FALSE**

$z = (3 - 0)/\sqrt{(1^2 + 1^2)} = 3/1.41 = 2.12$, $p \approx 0.03$. 95% CIs can overlap by half an arm and still give $p < 0.05$.

**2** The 95% CI for the difference B – A is [0.23, 5.77], which excludes 0. **TRUE**

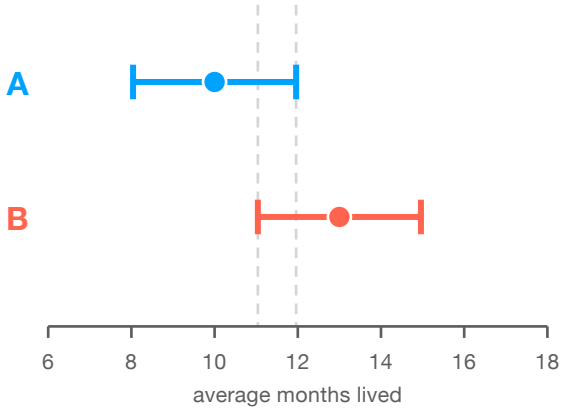
$3.0 \pm 1.96 \times \sqrt{(1^2 + 1^2)} = 3.0 \pm 2.77$. Test the difference with the difference's SE.

**3** If two 95% CIs don't overlap at all, the difference is significant. **TRUE**

With equal SEs, CIs that just touch give $p \approx 0.006$. Sufficient, but far from necessary.

# Where it goes wrong

± 95% confidence intervals

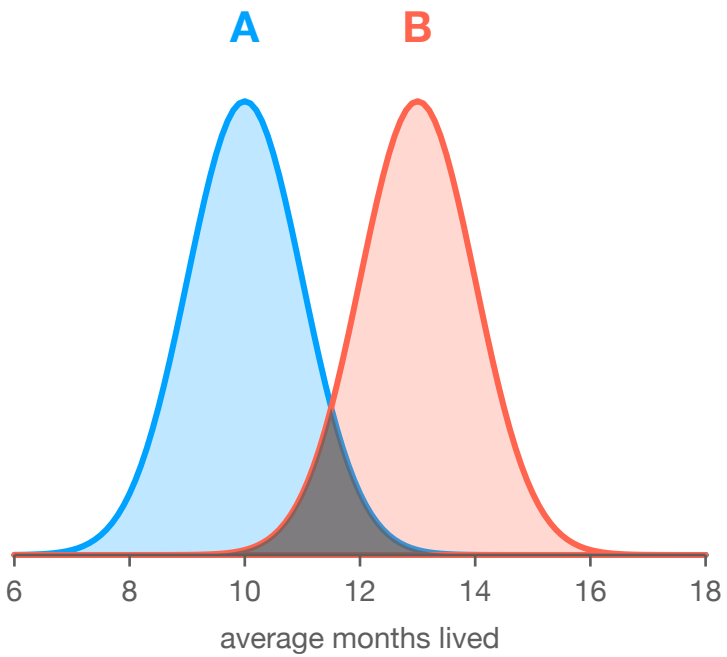



**1. “The CIs overlap, so the difference isn’t significant.”**

**Drug approval:** A review sees these overlapping CIs and calls the new drug no better than the standard one, so patients keep getting the drug that, by the right test ($p \approx 0.03$), gives them less time.

# Overlapping sampling distributions

Same drug trial (A = 10, B = 13 months, SE = 1). Gray: where the curves overlap.




## Same data, new picture

The same drug trial: average survival 10 months on A and 13 on B, SE = 1 each.

## Thought experiment

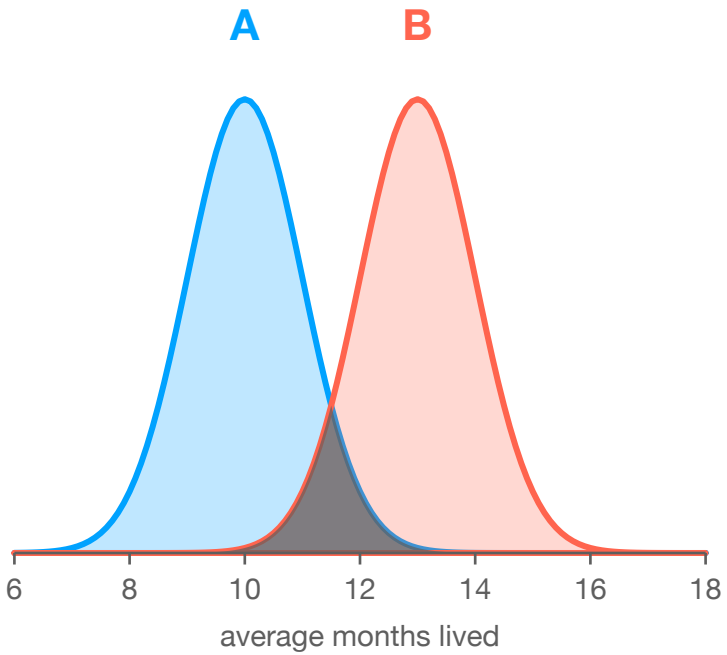
Re-run the trial with 25 new patients per drug, over and over. Each curve shows where that drug's average survival would land: centered on the observed average, with spread SE = 1 month.

## The shading

Gray marks averages either drug could plausibly produce.

# Overlapping sampling distributions

Same drug trial (A = 10, B = 13 months, SE = 1). Gray: where the curves overlap.




1 The curves overlap, so we can't say the drugs differ. T / F

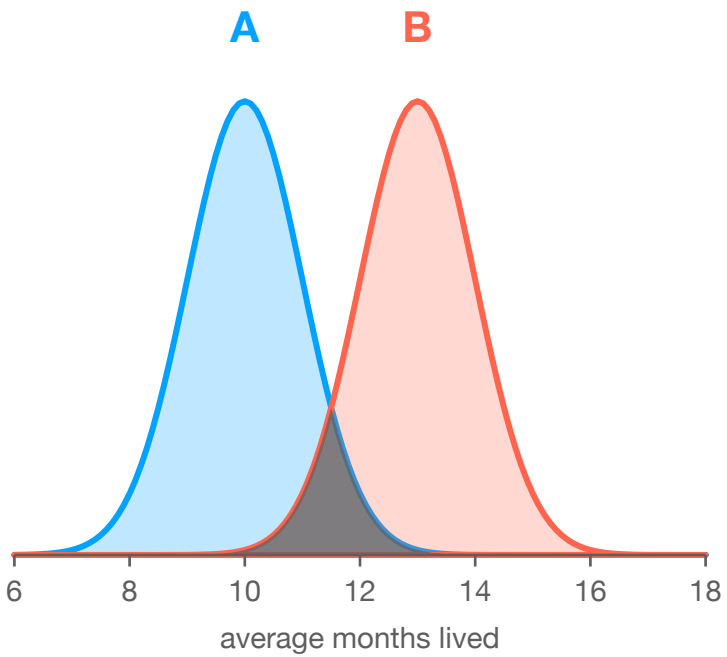
2 The gray overlap area is the p-value. T / F

3 The curves show where each drug's average survival would land, not how long individual patients live. T / F

# Overlapping sampling distributions

Answers

Same drug trial (A = 10, B = 13 months, SE = 1). Gray: where the curves overlap.




**1** The curves overlap, so we can't say the drugs differ. **FALSE**

Same data as the CI slides: p ≈ 0.03. Overlap of the two curves is not the test.

**2** The gray overlap area is the p-value. **FALSE**

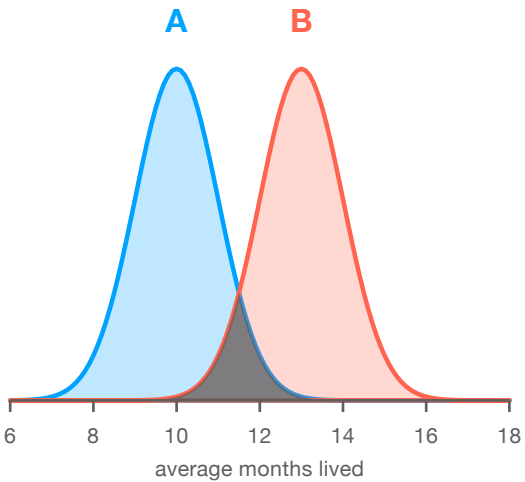
The overlap area is $2\Phi(-1.5) \approx 0.13$; p ≈ 0.03. The p-value comes from the distribution of the difference.

**3** The curves show where each drug's average survival would land, not how long individual patients live. **TRUE**

Their spread is SE = SD/√n = 1 month, while individual patients' survival spreads with SD = 5.

# Where it goes wrong






**1. “The curves overlap, so we can’t say the drugs differ.”**

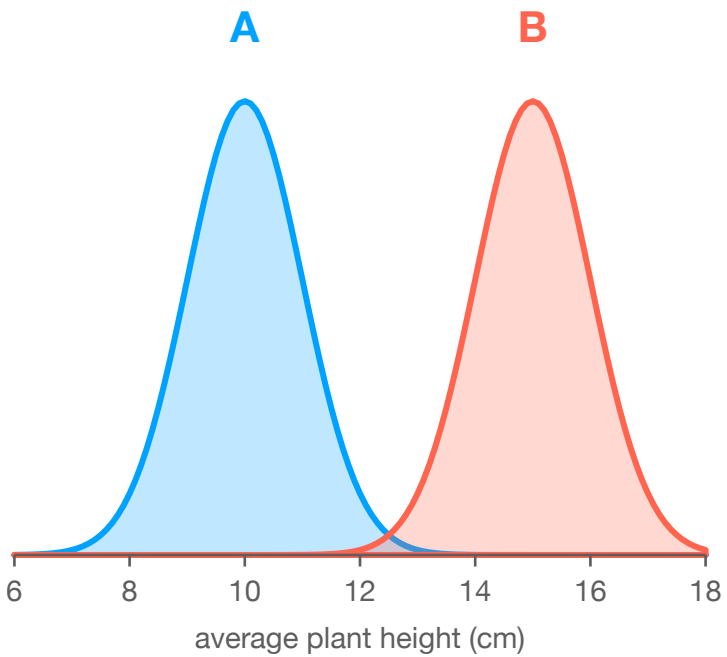
**Evaluating a program:** A school board sees the old and new curricula’s curves overlap and declares the new one useless. Overlap is the wrong test, and even a truly non-significant result means “not shown to differ”, not “shown to be the same”.

**2. “The gray overlap area is the p-value.”**

**Reporting:** An analyst reports “p = 0.13, not significant” from the overlap area and the project is shelved, when p ≈ 0.03. Computing p from the wrong distribution can flip a decision in either direction.

# Barely overlapping sampling distributions

Fertilizer A vs B: average height of 25 plants each, SE = 1 cm.




## The question

Does fertilizer B grow taller plants than fertilizer A?

## The data

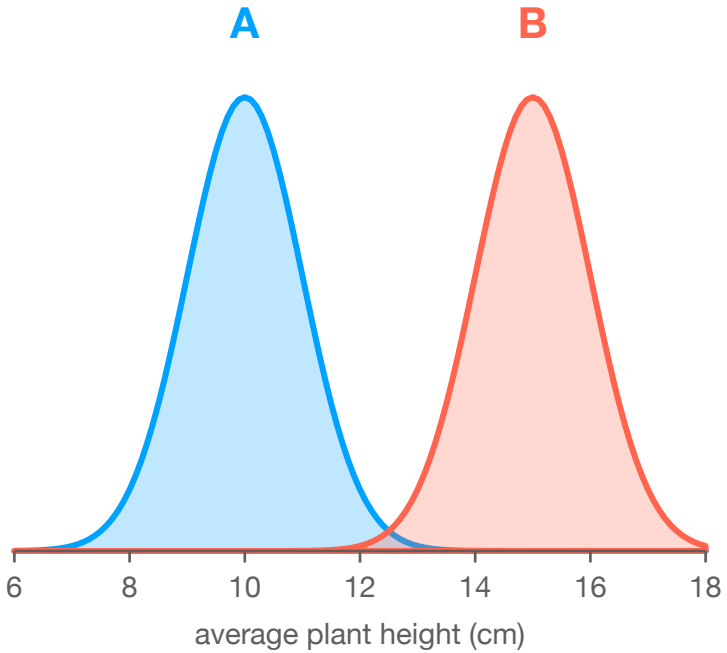
25 plants per fertilizer. A's plants average 10 cm and B's 15 cm. Single plants vary a lot: SD = 5 cm.

## The plot

Each curve shows where that fertilizer's average height would land if you grew 25 new plants: spread SE = $5/\sqrt{25} = 1$ cm.

# Barely overlapping sampling distributions

Fertilizer A vs B: average height of 25 plants each, SE = 1 cm.




1 The difference is significant. T / F

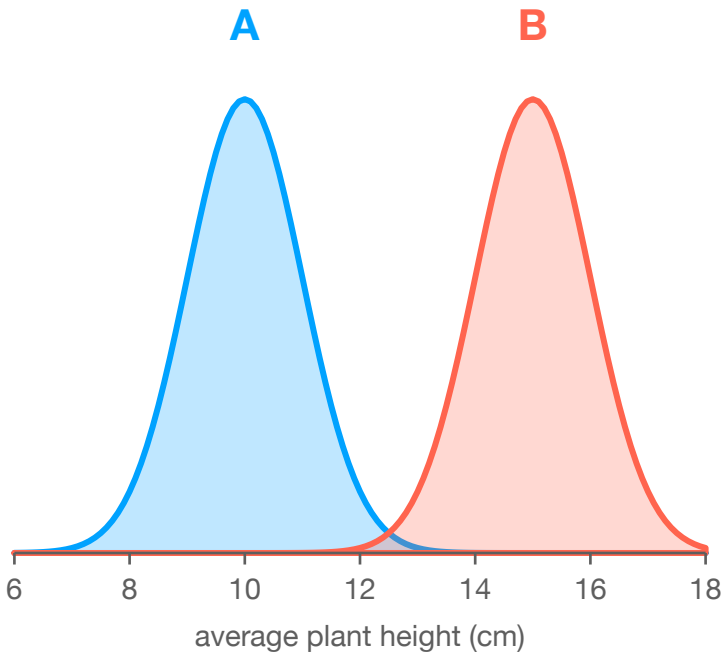
2 A random B plant is almost certainly taller than a random A plant. T / F

3 With 4× as many plants per group, each curve would be half as wide. T / F

# Barely overlapping sampling distributions

Answers

Fertilizer A vs B: average height of 25 plants each, SE = 1 cm.




**1** The difference is significant. **TRUE**

$z = (5 - 0)/\sqrt{(1^2 + 1^2)} = 5/1.41 = 3.54$, $p \approx 0.0004$.

**2** A random B plant is almost certainly taller than a random A plant. **FALSE**

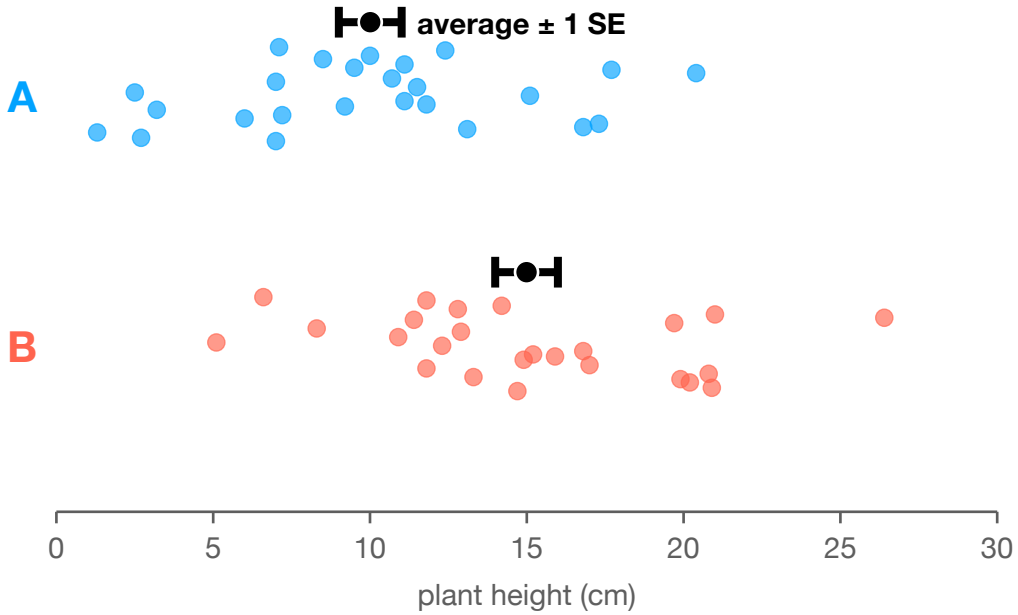
Single plants have SD 5, not 1, so P(B plant taller) = $\Phi(5/\sqrt{50}) \approx 0.76$.

**3** With 4× as many plants per group, each curve would be half as wide. **TRUE**

SE = SD/$\sqrt{n}$, and $\sqrt{4} = 2$.

# The data behind the plot

**Barely overlapping sampling distributions**




## Averages

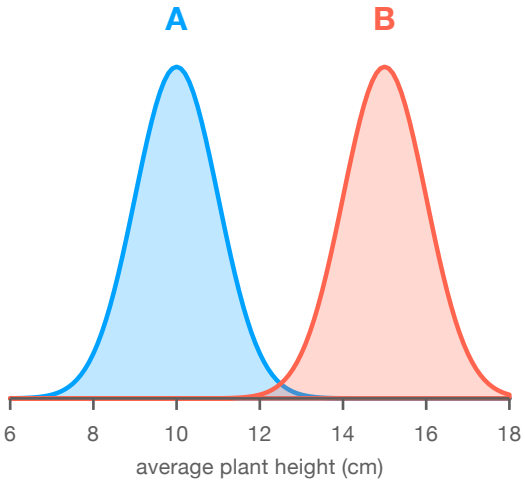
Clearly apart: the black bars don’t come close.

## Single plants

Overlap heavily. A random B plant beats a random A plant only about 76% of the time.

# Where it goes wrong

Barely overlapping sampling distributions

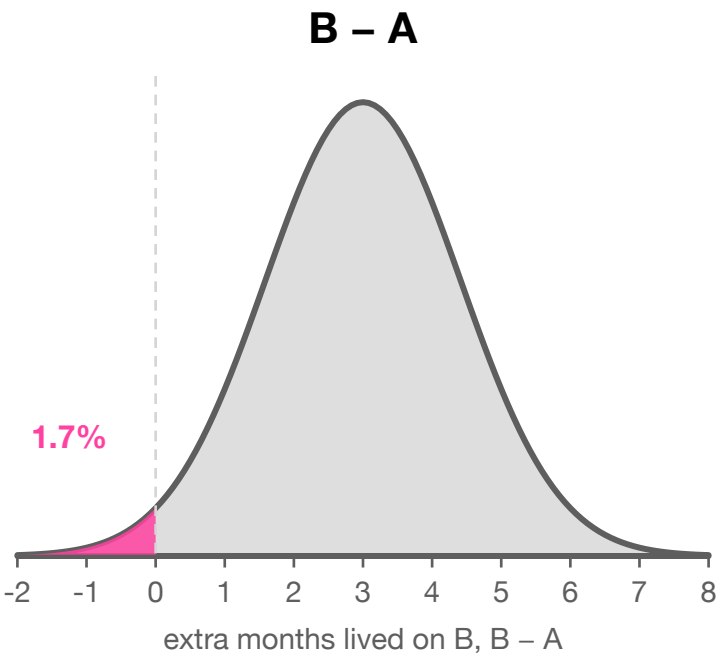



**2. “A random B plant is almost certainly taller than a random A plant.”**

**Treating individuals by group averages:** A drug lowers average blood pressure by a highly significant amount, and a clinic assumes every patient benefits. Hiring or admitting people by group averages makes the same mistake: a clear difference in means can coexist with heavily overlapping individuals.

# Sampling distribution of the difference

Same drug trial (A = 10, B = 13 months). Curve: the gap B – A. Pink: below 0.




## Same data, one curve

The same drug trial. What we care about is the gap: B – A = 13 – 10 = 3 extra months.

## Thought experiment

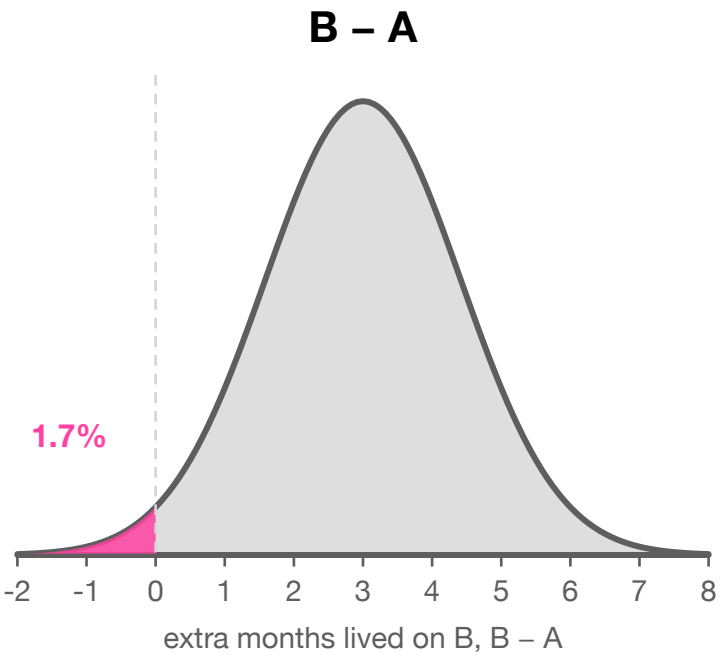
Re-run the trial many times and record the gap each time. The curve shows where it would land, centered at 3. How wide is it? That's statement 1.

## The shading

If the true gap really were 3, the pink area is the share of re-runs that would come out below 0: 1.7%.

# Sampling distribution of the difference

Same drug trial (A = 10, B = 13 months). Curve: the gap B – A. Pink: below 0.




**1** The SE of the gap is 1 + 1 = 2. T / F

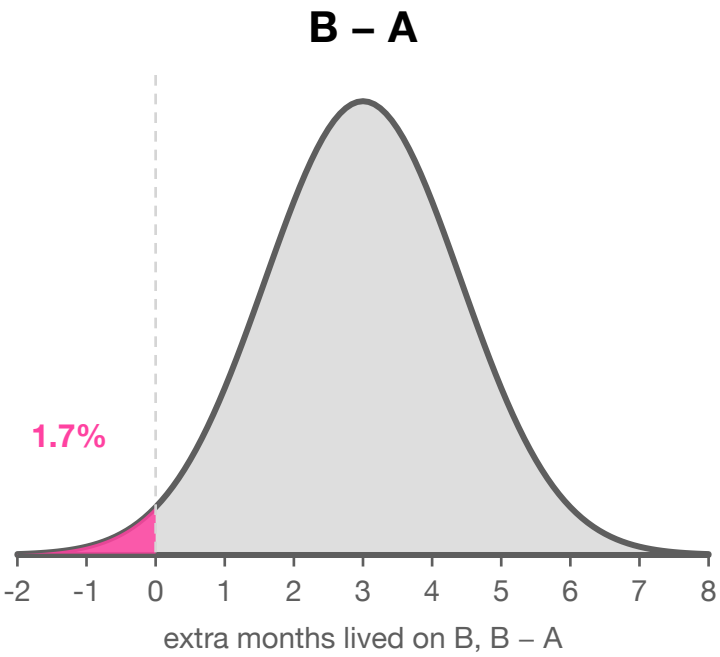
**2** There's a 1.7% chance the new drug is truly worse than the standard one. T / F

**3** If patients were matched in pairs (one on each drug), the two SEs alone wouldn't give this curve's width. T / F

# Sampling distribution of the difference

Answers

Same drug trial (A = 10, B = 13 months). Curve: the gap B – A. Pink: below 0.




**1** The SE of the gap is 1 + 1 = 2. **FALSE**

For independent averages the variances add, not the SEs: $\sqrt{(1^2 + 1^2)} = 1.41$.

**2** There's a 1.7% chance the new drug is truly worse than the standard one. **FALSE**

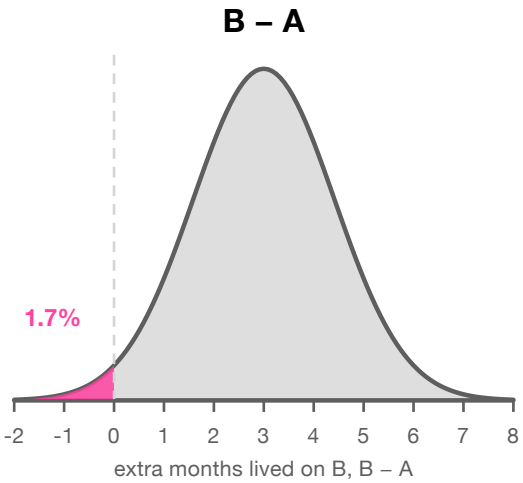
To a frequentist the true averages are fixed. A Bayesian with a flat prior would say exactly this (Efron 1986).

**3** If patients were matched in pairs (one on each drug), the two SEs alone wouldn't give this curve's width. **TRUE**

Var(B – A) = Var(A) + Var(B) – 2Cov(A, B). The SEs don't show the covariance.

# Where it goes wrong

Sampling distribution of the difference




**1. “The SE of the gap is 1 + 1 = 2.”**

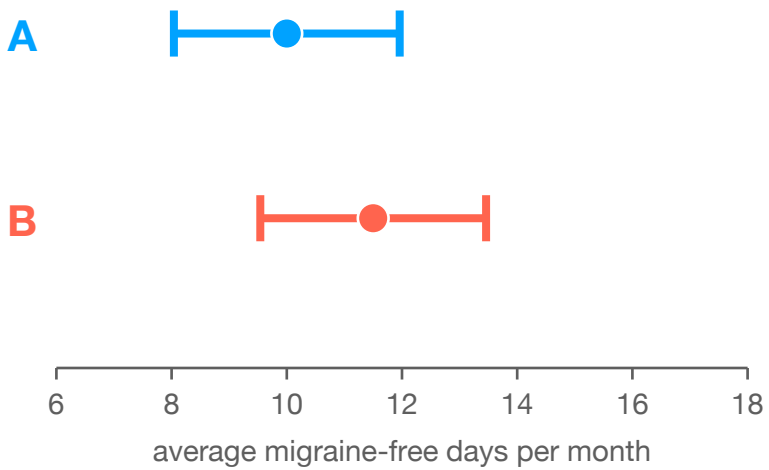
**Wasted studies:** Adding SEs overstates the uncertainty in a difference. A team that plans its sample size this way recruits about twice as many people as it needs, or writes off a real difference as noise.

**2. “There’s a 1.7% chance the new drug is truly worse than the standard one.”**

**Prosecutor’s fallacy:** In the Sally Clark trial, “1 in 73 million” (the claimed chance of two SIDS deaths in one family, itself invalid because it assumed independent deaths) was heard as the chance she was innocent. Reading p = 0.03 as “3% chance of no effect” makes the same swap. If 10% of tested hypotheses are true, about 36% of significant results are false positives (80% power).

# Overlapping CIs, paired data

The same 30 patients on placebo (A) and on the drug (B).
Bars: 95% CI.




## The question

Does a migraine drug give patients more migraine-free days?

## The data

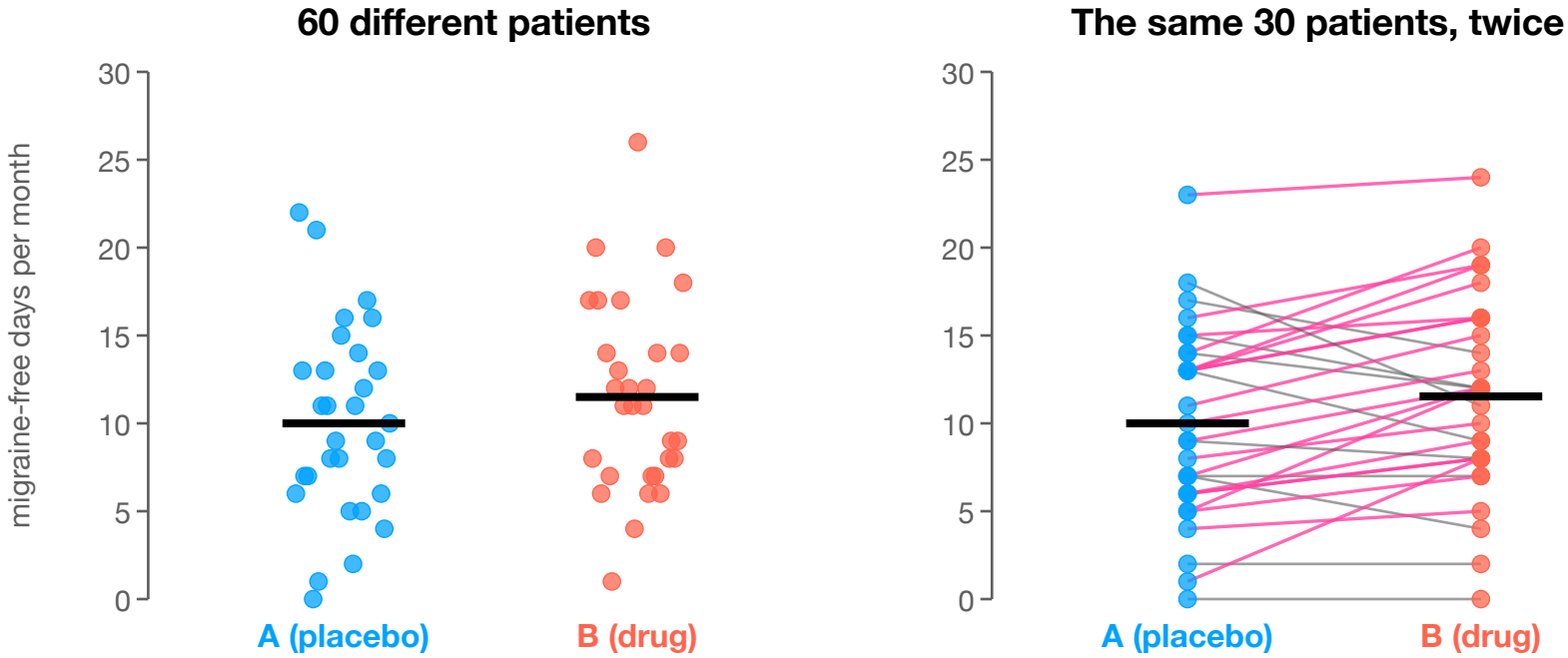
The same 30 patients spend one month on a placebo (A) and one month on the drug (B). They average 10 migraine-free days on placebo and 11.5 on the drug; each SE = 1.

## The catch

Patients with frequent migraines have few good days both months, and patients with rare migraines have many, so each patient's two counts are strongly correlated (r = 0.8).

# The data behind the plot

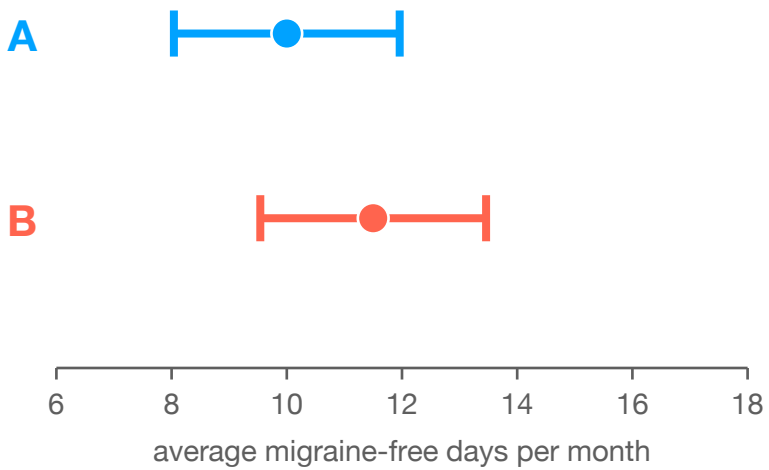
Overlapping CIs, paired data




**Same averages (10 vs 11.5) and same SEs on both sides.** Black bars mark the averages. On the right, each line is one patient; pink lines went up. 20 of 30 patients had more good days on the drug, 7 had fewer and 3 were unchanged.

# Overlapping CIs, paired data

The same 30 patients on placebo (A) and on the drug (B).
Bars: 95% CI.




1 The CIs overlap a lot, so the drug made no significant difference. T / F

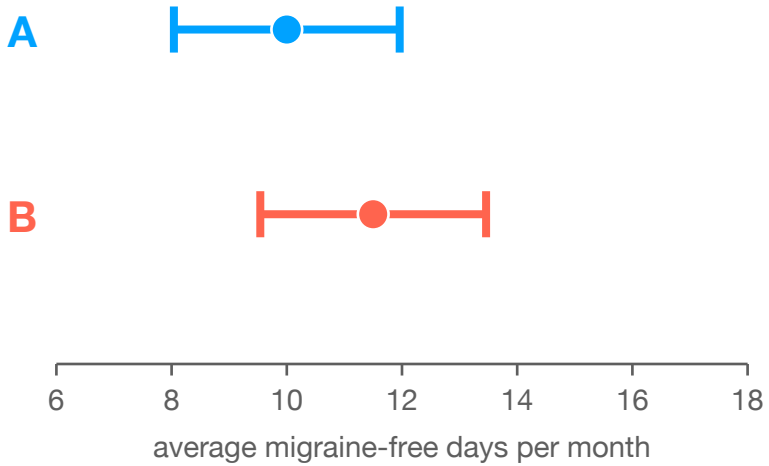
2 To judge the drug's effect, you need to know how correlated each patient's two counts are. T / F

3 Each CI is wrong because the same patients appear in both. T / F

# Overlapping CIs, paired data

## Answers

The same 30 patients on placebo (A) and on the drug (B). Bars: 95% CI.




**1** The CIs overlap a lot, so the drug made no significant difference.

**FALSE**

$z = (1.5 - 0)/\sqrt{(1^2 + 1^2 - 2 \cdot 0.8)} = 1.5/0.63 = 2.37$, $p \approx 0.02$.

**2** To judge the drug's effect, you need to know how correlated each patient's two counts are.

**TRUE**

Overlap rules of thumb assume independent groups.

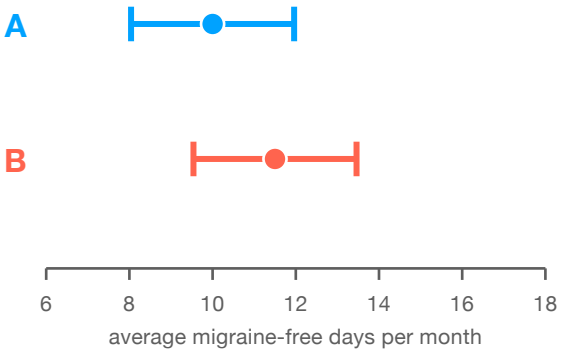
**3** Each CI is wrong because the same patients appear in both.

**FALSE**

Each CI is valid for its own average. It just isn't the tool for comparing them.

# Where it goes wrong






**1. “The CIs overlap a lot, so the drug made no significant difference.”**

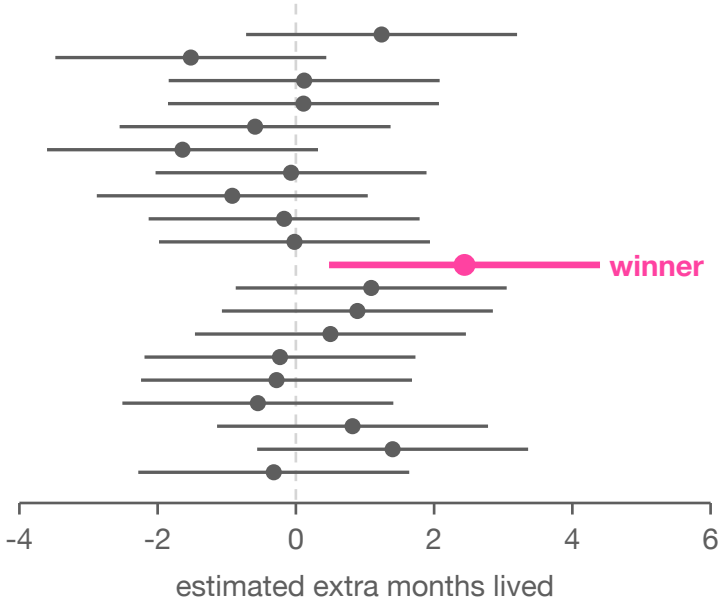
**Crossover trials:** A reviewer sees the placebo and drug CIs overlap and calls the migraine drug ineffective, so it’s shelved. Each patient is their own control; the paired test ($p \approx 0.02$) shows a reliable gain that the overlap check throws away.

**3. “Each CI is wrong because the same patients appear in both.”**

**Over-correcting:** A reviewer strikes valid per-condition CIs as “invalid for paired data”, and readers lose the size and precision of each mean. Keep them, and add a CI for the paired difference.

# Picking the best of 20 drugs

20 candidate drugs, each on its own small group of patients. Effect: extra months lived vs standard care, SE = 1 each. Bars: 95% CI.




## The question

Which of 20 candidate drugs should go on to a large trial?

## The data

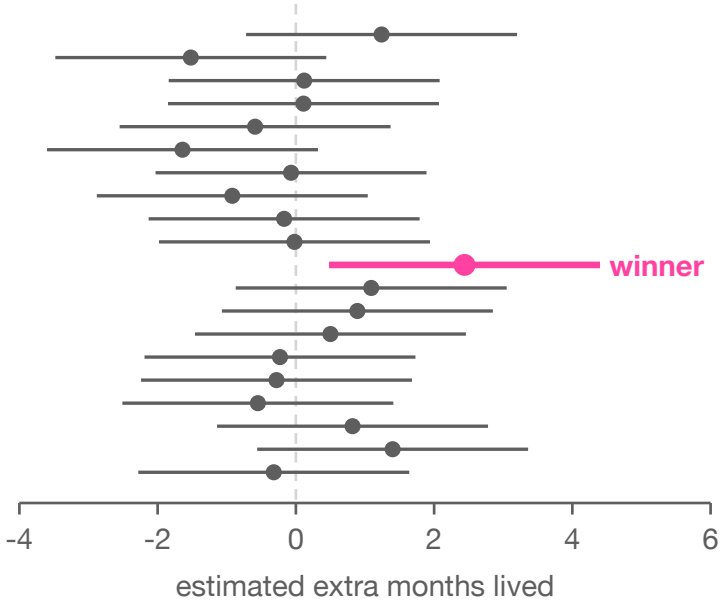
Each candidate is given to its own small group of patients. Its effect is extra months lived compared with standard care, with SE = 1.

## The rule

Advance the candidate with the biggest effect if its 95% CI excludes 0. From experience, most candidates do nothing.

# Picking the best of 20 drugs

20 candidate drugs, each on its own small group of patients. Effect: extra months lived vs standard care, SE = 1 each. Bars: 95% CI.




1 Each interval comes from a method that captures the true effect 95% of the time. T / F

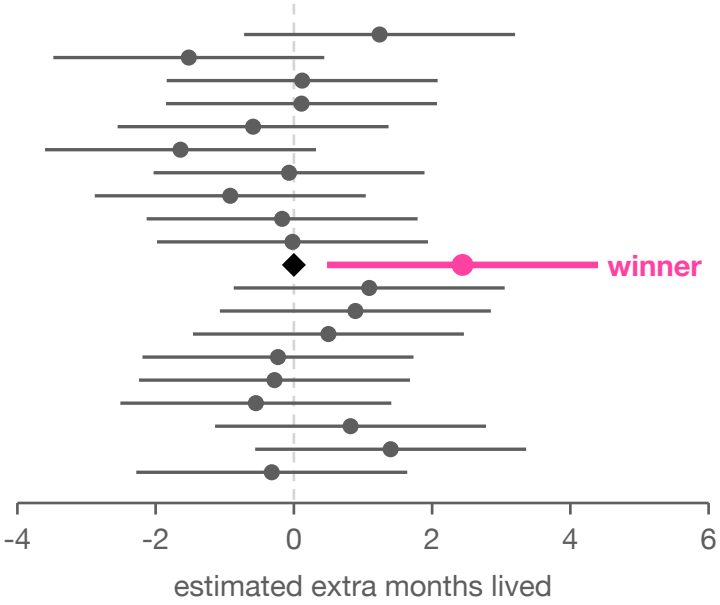
2 There's a 95% chance the winner's true effect is between 0.48 and 4.40 months. T / F

3 The winner's 2.44 extra months is a good guess of what the large trial will show. T / F

# Picking the best of 20 drugs

## Answers

20 candidate drugs, each on its own small group of patients. Effect: extra months lived vs standard care, SE = 1 each. Bars: 95% CI.




**1** Each interval comes from a method that captures the true effect 95% of the time. **TRUE**

Each CI is a correct 95% procedure. In simulation, 95.0% of all intervals contain their true effect.

**2** There's a 95% chance the winner's true effect is between 0.48 and 4.40 months. **FALSE**

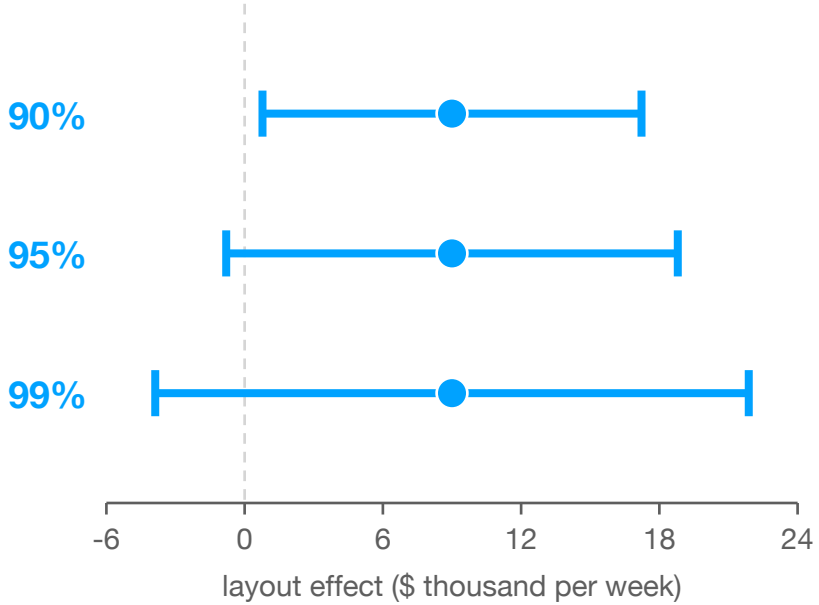
We're looking at it because it won, and most candidates do nothing. In 20,000 simulated screens, only 36% of the winners' CIs contained the true effect.

**3** The winner's 2.44 extra months is a good guess of what the large trial will show. **FALSE**

Winner's curse: the winners' estimates averaged 2.5 months, their true effects 0.45. 55% truly did nothing.

# Reporting 90%, 95% and 99% CIs

One regression coefficient: estimate 9, SE 5. Bars: 90%, 95% and 99% CIs.




## The question

Does a new store layout raise weekly sales? A regression gives the layout's coefficient: 9 (thousand dollars a week), SE = 5.

## The plot

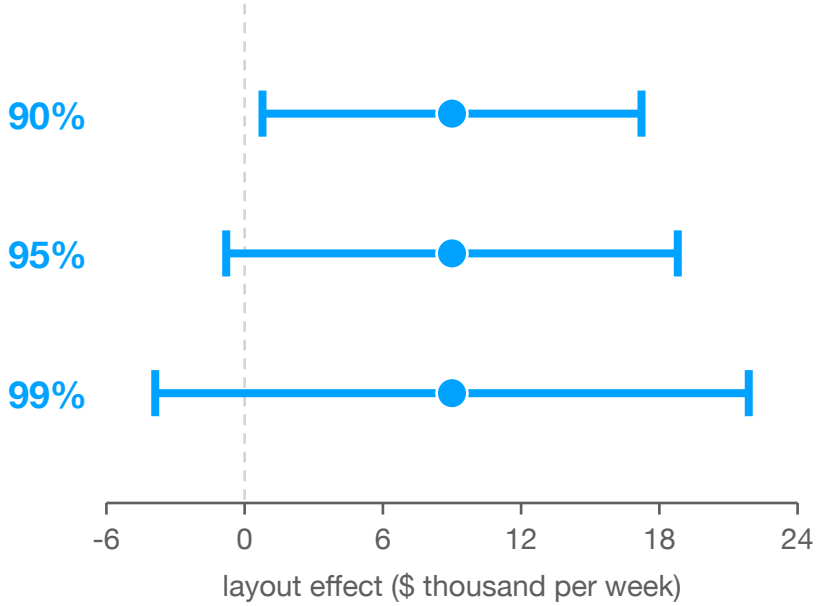
One estimate, three intervals: 9 ± 1.645, 1.96 and 2.576 SE. The 90% CI excludes 0; the 95% and 99% CIs don't.

## p-hacking

Trying analyses (outcomes, subsets, covariates, stopping points) until one crosses the bar, then reporting only that one.

# Reporting 90%, 95% and 99% CIs

One regression coefficient: estimate 9, SE 5. Bars: 90%, 95% and 99% CIs.




1 Reporting all three intervals gives three chances at a significant result. T / F

2 If the effect is 0, P(any interval excludes 0) = 1 − .90·.95·.99 ≈ 15%. T / F

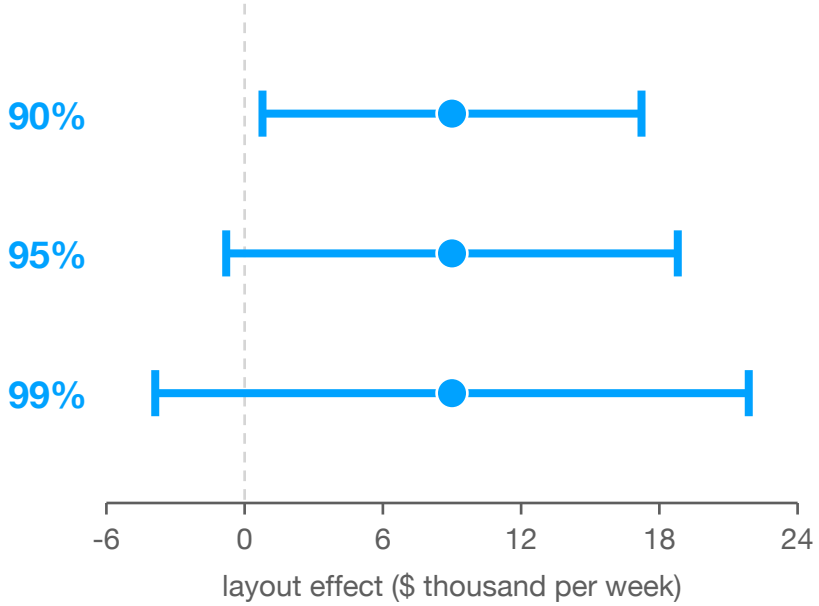
3 Calling it significant at 90%, after seeing 95% fail, is p-hacking. T / F

4 Reporting only the best of 5 outcomes raises the false-positive rate to about 23%. T / F

# Reporting 90%, 95% and 99% CIs

## Answers

One regression coefficient: estimate 9, SE 5. Bars: 90%, 95% and 99% CIs.




**1** Reporting all three intervals gives three chances at a significant result. **FALSE**

One estimate, one SE: $z = (9 - 0)/5 = 1.8$, $p \approx 0.07$. Three descriptions of one fact.

**2** If the effect is 0, P(any interval excludes 0) $= 1 - .90 \cdot .95 \cdot .99 \approx 15\%$. **FALSE**

The intervals are nested, so it's just the 90% one: exactly 10%.

**3** Calling it significant at 90%, after seeing 95% fail, is p-hacking. **TRUE**

That's choosing $\alpha$ after the data. Report $p \approx 0.07$ instead.

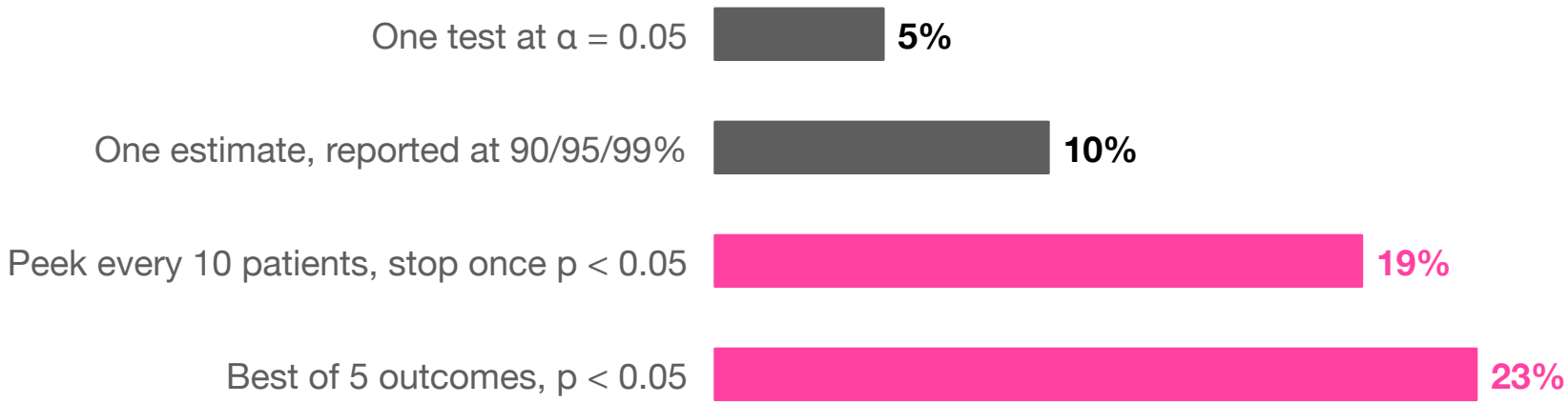
**4** Reporting only the best of 5 outcomes raises the false-positive rate to about 23%. **TRUE**

Five new chances: $1 - 0.95^5 \approx 0.23$.

# Descriptions vs. new chances

Reporting 90%, 95% and 99% CIs

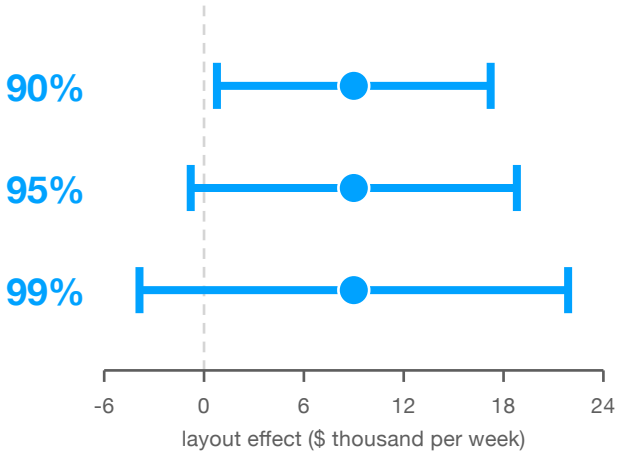
**False-positive rate when the true effect is 0**




**Gray: descriptions of one result.** Reporting several levels can't push the error rate above the loosest level you claim. **Pink: new chances.** Each extra try is another roll of the dice, and the reported p-value no longer means what it says.

# Where it goes wrong






**1. “Reporting all three intervals gives three chances at a significant result.”**

**Misplaced suspicion:** A reviewer calls a table of 90/95/99% intervals “p-hacked” and asks for one level only. Readers lose information, and the question that matters (how many analyses were tried before this one) goes unasked.

**2. “If the effect is 0, P(any interval excludes 0) = 1 – .90·.95·.99 ≈ 15%.”**

**Over-correcting:** An analyst applies a multiple-comparisons correction to the three confidence levels as if they were three tests, and a real result is thrown out. Corrections are for separate chances, not separate descriptions.

# Rules of thumb

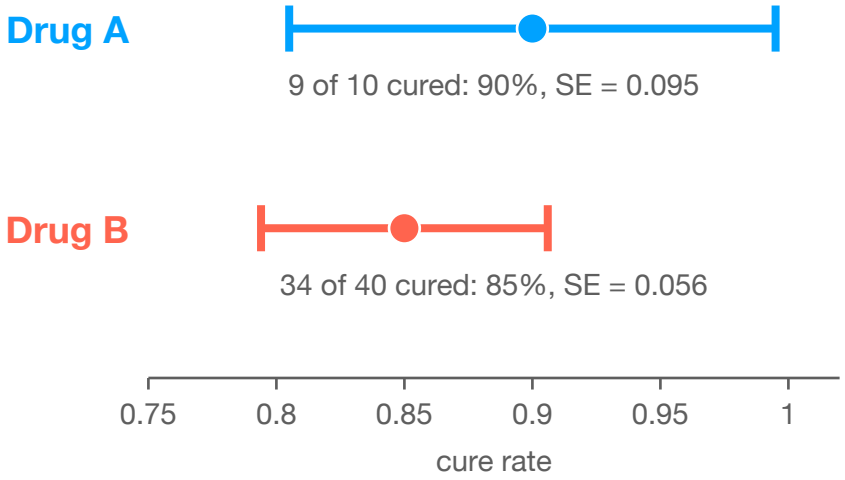
Independent groups with similar SEs, $\alpha = 0.05$

| $p \approx 0.16$ | $p \approx 0.04$ | $p \approx 0.006$ | ? |
|---|---|---|---|
| ±1 SE bars just touch | 95% CIs overlap by half an arm | 95% CIs just touch | Paired data: overlap tells you nothing |

**Comparing two estimates?** Test the difference. Its SE is $\sqrt{(SE_A^2 + SE_B^2)}$ for independent estimates, and needs the covariance for paired ones.

# Does inference ever flip the answer?

Cure rates ± 1 SE, with SE = $\sqrt{(\hat{p}(1-\hat{p})/n)}$.




## The question

You can fund one drug for a $100M trial. Drug A cured 9 of 10 patients; drug B cured 34 of 40.

## Think · 2 min

Which drug do you pursue, and how sure are you? The point estimate says A. Is that the right call?

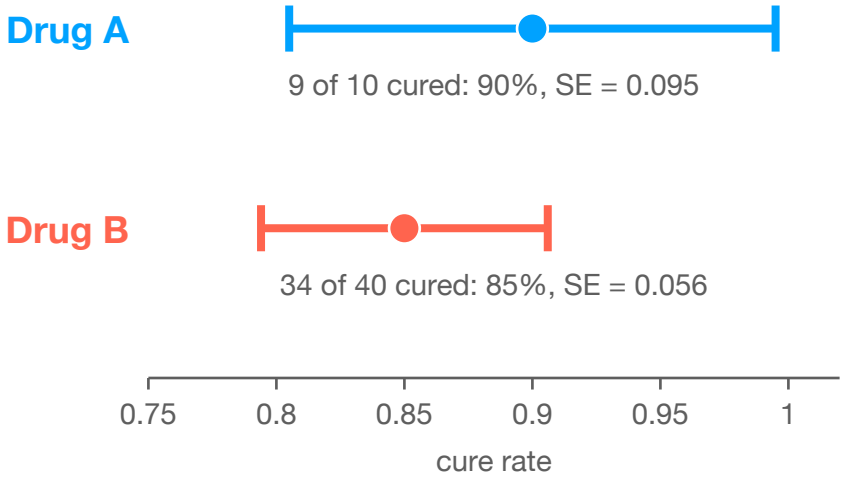
## Then

If inference never changes which estimate is bigger, why do it at all?

# Does inference ever flip the answer?

Answers

Cure rates ± 1 SE, with SE = $\sqrt{\hat{p}(1-\hat{p})/n}$.




### The lead is noise

A leads by 5 points, but the gap's SE is 0.11: z = (0.05 – 0)/0.11 = 0.45, p ≈ 0.65. Gaps this big are common even if the drugs are equal.

### "Don't know" can flip next time

Route B led by 2 minutes (p ≈ 0.16). Rerun that study and A comes out ahead about 16% of the time.

### The winner's payoff shrinks

Winning drugs from the screen promised 2.5 extra months and delivered 0.45.

### Inference tells you how much to bet

Pursue, collect more data, or check the break-even. The energy pilot estimated $400 saved, but its CI reached down to $122, against a $200 cost.